

PRERIT SANJAY MITTAL

Agentic AI · Edge Inference · MLOps · Reinforcement Learning

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SUMMARY

AI/ML Engineer with production ML and data systems experience at GE Vernova, now focused on MLOps and agentic AI engineering. Background spans computer vision, NLP, and LLM fine-tuning (LoRA/GRPO), with hands-on experience shipping Dockerized FastAPI inference services and multi-agent workflows, from RAG pipelines to LangGraph-based agent systems orchestrating custom MCP tool servers. Fluent with AI-native dev tools including Claude Code and Cursor. Strong bias toward working prototypes shipped fast over perfect plans.

EDUCATION

Rochester Institute of Technology, Rochester, NY

Aug. 2024 – May 2026

Master of Science in Computer Engineering; GPA: 3.97/4

Indian Institute of Information Technology, Lucknow, Lucknow, India

Aug. 2019 – May 2023

Bachelor of Technology in Computer Science; GPA: 3.56/4

SKILLS

Languages & Tools: Python, C++, CUDA, SQL, Git, Bash, Docker, Kubernetes

Frameworks & Libraries: PyTorch, TensorFlow, Keras, NumPy, Pandas, Scikit-learn, OpenCV, Flask, FastAPI

Cloud & MLOps: Weights & Biases (W&B), DVC, MLflow, GitHub Actions, AWS (EC2, S3, SageMaker, Bedrock)

DL & ML Techniques: CNNs, NLP, Reinforcement Learning, LLMs, Model Optimization, Inference Acceleration

Data Engineering & Visualization: Databricks, PySpark, MongoDB, MSSQL, PostgreSQL, Neo4j, Terraform

Agentic AI & RAG: LangGraph, LangChain, RAG, MCP, Tool Calling, Hybrid Retrieval, BM25, Vector Databases

PROFESSIONAL EXPERIENCE

Graduate Research Assistant

Aug. 2025 – May 2026

Rochester Institute of Technology

Rochester, NY

- Architected a **dual-model edge-to-cloud perception system** via Triton Server, serving quantized YOLOv8-seg and Depth Anything V2, achieving **72 ms median** end-to-end latency and **98 ms p95**.
- Enforced **codified latency SLAs** ($p95 < 120 \text{ ms}$) by building an offline benchmarking tool for per-model p95/p99 inference analysis, catching regressions before deployment.
- Improved **readiness for real-time obstacle avoidance**, maintaining a **~9.8 Hz** usable update rate with **1.4% stale frames**, by streaming near/mid/far obstacle metadata over a self-hosted WebRTC DataChannel.
- Led multimodal dataset collection for a custom 3-DOF robotic arm in LeRobot format, handing off the pipeline for future **VLA (Vision-Language-Action)** model fine-tuning.

Software Engineer

Jan. 2023 – Aug. 2024

GE Vernova

Bangalore, India

- Developed an **Advanced Process Control (APC) digital twin** using PCA and neural networks to optimize temperature & pressure setpoints, reducing process variability by **15%** and increasing throughput by **10%**.
- Automated baseline checks post-upgrade, cutting manual review time from **42 hours to 1.5 hours**, achieving **95%** defect detection accuracy, and saving approximately **\$5K** per upgrade cycle.
- Streamlined **database migration**, cutting transfer time for 80K+ records from **1 hour to 15 minutes** by automating validation and transformation, improving consistency and throughput.

PROJECTS

Agentic Supply Chain Forecasting Pipeline | LangGraph · MCP · dbt · PostgreSQL · Databricks DLT

- Engineered a self-correcting four-agent LangGraph pipeline over hierarchical Walmart retail sales series, achieving a **92.4% defect catch rate** at 0.8% false-quarantine rate and **0.584 WRMSSE** forecast accuracy.
- Orchestrated tool-based agent access via three custom MCP servers atop a dual-backend abstraction supporting dbt-modeled Postgres and Databricks DLT (Auto Loader over Unity Catalog).

Adaptive Multi-Agent RAG System | *Agent Swarm · ChromaDB · Qdrant · BM25 · RAGAS*

- Engineered an uncertainty-aware multi-agent RAG system, achieving **0.89 RAGAS faithfulness**, **0.86 precision@3**, and a **4.3% hallucination rate**, by orchestrating Agent Swarm with adaptive retrieval.
- Reduced weak-context hallucinations by **77%** by implementing cosine-similarity uncertainty scoring, threshold-based refusal, claim-level self-correction loops, and benchmarking search and retrieval tools.

Wafer Quality Checker – Production MLOps Pipeline | *MLflow · DVC · GitHub Actions · AWS*

- Built a production-grade wafer quality prediction system for SECOM sensor data, enabling auditable Pass/Fail classification through automated validation, MLflow tracking, and inference logging for drift monitoring.
- Accelerated retraining by **60–80%** by versioning feature snapshots with DVC, automating GitHub Actions CI/CD, and launching on-demand EC2 training pipelines.

CUDA Optimization for VGG16 | *CUDA · cuBLAS · Nsight Compute · ONNX Runtime*

- Redesigned VGG16 convolution kernels on an NVIDIA A16 GPU using naïve, shared-memory-tiled, and register-coarsened implementations; validated outputs against ONNX Runtime across all 13 layers.
- Achieved a **2.5× throughput gain**, **30%** higher GPU utilization, and **15%** lower memory latency, profiling kernels with Nsight Compute to guide optimization via hierarchical tiling and im2col + cuBLAS integration.

Adaptive LLM Fine-Tuning with GRPO | *LoRA · GRPO · Unsloth · Quantization*

- Engineered a modular fine-tuning framework using LoRA adapters, Unsloth, and Group Relative Policy Optimization (GRPO) to enhance mathematical reasoning on the GSM8K benchmark.
- Improved GSM8K accuracy by **8.5%** across Qwen2.5-3B, Gemma 1B, and Llama 3.1-8B by implementing a reward-based feedback loop with KL penalty and advantage normalization and quantized fine-tuning.

RESEARCH EXPERIENCE

Noise-Aware RL for Quantum Circuit Routing

Sep. 2025 – Dec. 2025

Rochester Institute of Technology

Rochester, NY

- Extended a **PPO-based RL routing policy** (TwisteRL, Rust/Python hybrid) with a **noise-aware heuristic**, modeling accumulated per-edge gate error as a hashmap over the device coupling graph to bias routing against error-prone qubit links.
- Tuned the noise penalty per CNOT operation through empirical testing, balancing signal strength so noise avoidance influenced routing decisions without overriding the base reward's depth/gate-count optimization.
- Validated the policy on a cyclic 5-qubit ring, showing the noise-aware model took a longer but noise-avoidant path around a loaded edge, and produced a **more even CNOT distribution** across edges than the non-noise-aware baseline over 10 repetitions.

Explainable Alzheimer's Disease Detection from Speech

Jun. 2025 – Aug. 2025

Rochester Institute of Technology

Rochester, NY

- Trained an **XGBoost classifier on eGeMAPS acoustic features** extracted from the ADReSS speech dataset to detect Alzheimer's Disease, addressing a minimum-segment-length constraint by shifting feature extraction from phoneme-level to word-level framing.
- Tuned PDSM (Phoneme-Discretized Saliency Map) hyperparameters via faithfulness scoring, lifting faithfulness from **−0.004 (GradSHAP)** to **0.033** on AD-positive samples.
- Ran **Cohen's d** and **TreeSHAP analysis on eGeMAPS** features, identifying loudness, spectral flux, and voice quality as the top acoustic markers of AD speech.

LEADERSHIP AND ACADEMIC SERVICE

Graduate Teaching Assistant – Reinforcement Learning: Facilitated lab sessions for 50+ students on Markov Decision Processes (MDPs) and Deep Q-Networks (DQN).

Graduate Delegate & Teacher's Choice Award Recipient: Elected representative for the Computer Engineering cohort and recognized by faculty for collaborative leadership.

RIT Research Symposium – 1st Place Poster Award: Presented a monocular depth-based obstacle avoidance system for autonomous UAV navigation.